

low by around 20 per cent.

What are the improvements being made in energy storage systems? There are talks about battery alternatives like ultra-capacitors and flywheel.

Ultracapacitors are currently being used in hybrid (electric+diesel) cars and wind and solar power systems. When you apply brakes to a speeding car, a lot of energy is generated, which is fed to the capacitor. Ultracapacitors charge almost instantly, while batteries require 8-10 hours for charging. Also, the life of an ultracapacitor is more than 25 years, compared to generally not more than three years of a battery. So ultracapacitors are being used not only for the automotive industry but also for wind and solar power generation.

But, whenever any XYZ technology is launched, the initial cost is high. For example, earlier, solar cells were available for Rs 250/watt, now they are available for Rs 80/watt. With the kind of requirement in India, the cost may come down to Rs 50/watt.

Similarly, currently, the initial cost of ultracapacitors is high. Once there are more manufacturers and more demand, the cost will automatically come down. Then the ultracapacitors could be used for UPS as well.

Flywheel UPS are rotary UPS that consist of a motor, generator and flywheel. When mains is available, the motor rotates the flywheel. When mains fails, the flywheel keeps rotating and generating power. In the meantime, the generator turns on and starts rotating the motor. So, practically, you have uninterrupted power.

Flywheel technology is very costly and used for critical power systems of more than 500 kVA.

When will we see solar backup systems for computers?

Solar-powered laptops, mobile phones and other devices are already available in the market. At present, these are not much popular because their cost is high. But with government subsidies and other support, by the end of 2012, lots of these will be used.

The government has asked all the banks to have some fixed number of ATMs powered by solar systems. It is also thinking about bringing the concept of grid-tied inverters. Under this concept, the user will install a solar system on the rooftop or any other place at his house. During the daytime, the extra power generated can be fed to the electricity grid and redrawn at night when you need it.

What are the key challenges for designing UPS systems for Indian conditions?

Every 50 km in India, you'll find different challenges. Some states have surplus power, while some states don't have power at all. Power cut in states like Delhi is only two hours a day, but 50 km from this place, i.e., in Faridabad or Gurgaon, it could be for six to eight hours

a day. You don't know the input voltage at your end. In India, the power has to be minimum 230V $\pm$ 10 per cent. But, in many areas, the input power is not even 120V at night, so you have to use double stabilisers so that you can operate your AC.

Also, the Indian environment is dusty and hot, which adds to the problem of heat generated by the system. Then, there is the issue of backup time required. When you buy any UPS from Europe, the standard backup time available is not more than five minutes. But here the backup requirement is sometimes up to eight hours.

Coming to end-user-oriented issues, how to select the battery for a UPS system?

There are various types of batteries: Automotive, tubular and SMF. Each has its own advantages and disadvantages. Some are low-cost. Others are expensive, but their life is also longer. Then there is the issue of gases. For AC environments, go for SMF batteries, not the automotive and tubular batteries.

To determine the load, ascertain the requirement. For example, for a room having only lighting load, count the total number of lights and the power consumed by each, then add that all to come to the total load. Similarly, if you want to connect an XYZ load, go through the technical manual of the supplier. Or, you can use a power meter to measure the current running load.

Is the UPS-cum-inverter reality?

There are two kinds of UPS systems: Online and line-interactive. Online systems are UPS-cum-inverter, because the inverter is 100 per cent time 'on' and the load is always through the inverter. In line-interactive systems, there is a changeover time.

Any system having changeover time of more than 10 ms is not a UPS system in true sense. For your home application, in case power fails, there is a changeover and you get power. For you, it's uninterrupted power. But, for office applications running on computers, with changeover, the systems reboot. So it's not uninterrupted power.

What are the other popular marketing gimmicks being used to confuse customers about UPS systems?

First, suppose you want a 5kVA UPS system. The vendor comes to you and studies that your load is not even 2 kVA. He can supply you a 3kVA system with a marking of 5kVA.

Second, you invite the vendor for discussion about a 5kVA system with 30-minute backup time. But he supplies a 5kVA system with batteries giving lower backup time.

Sometimes the buyer himself is not aware. His actual requirement is of a 20kVA UPS but places the order for 10kVA, going by the advice of his electrician or his own judgement. ●